Trainer v2 smoke test: results and interpretation

Job: `trainer_v2_smoke` (27998524)
Run date: 2026-10-29
Report date: 2026-10-30
Verdict: PASS for the trainer-v2 behavior and end-to-end smoke path. This is a functional check, not evidence that v2 improves affinity-prediction accuracy.

1. Executive summary

The job completed all three training arms, inspected the saved artifacts, and rescored the selected validation structures. The artifacts show the distinction the migration was intended to restore: v2 LoRA and full fine-tuning changed the affinity pairformer stacks and both affinity modules, while the legacy arm changed only `affinity_module1` heads. The legacy run also logged the expected no-gradient warnings for its frozen stack and out-of-forward targets.

Both vanilla and v2-LoRA rescoring produced 18 successful rows for the same 18 unique structures. The smoke script selected 20 validation rows, but two structure names were duplicated, so only 18 distinct PDB inputs were rescored. The mean absolute prediction difference on the paired successful rows was **0.564**; this establishes that the adapter affects scored outputs, not that those outputs are more accurate.

The main limitations are the very small, group-clustered subset; only two epochs per arm; no censored training examples; and a dirty checkout. Treat the results as confirmation of mechanics, not as a model comparison or a reproducibility record for a clean commit.

2. Test setup

Item	Observed configuration
GPU	NVIDIA H200 NVL, 143,771 MiB reported total memory
Training subset	64 rows from 2 assay groups
Validation subset	32 rows from 3 assay groups
Training duration	2 epochs per arm; batch size 8; 1 recycling step
Loss	`censored_intra_assay_huber`
v2 LoRA	Rank 8, alpha 16; default `affinity_heads_pairformer`; ensemble output
Legacy LoRA	Rank 8, alpha 16; `--legacy-trainer`; `heads_pairformer`
v2 full fine-tune	`affinity_heads_pairformer`; ensemble output; validation CSV supplied
Source revision	`1b7e973b592513231565ea9032cbce6691b1e7dc`, with dirty working tree

The run directory was `/scratch/maom_root/maom/limcaoco/trainer_v2_smoke_27998524`. The job header reported 20 modified tracked files; artifact provenance also records `boltz_git_dirty=true`. Consequently, the commit hash alone does not identify the exact code that ran.

3. Did the migration mitigation work?

The migration note describes a defect in the old reentrant activation-checkpoint path: when the frozen trunk supplied inputs without gradients, the checkpointed pairformer stack did not get an autograd graph. The v2 path uses non-reentrant checkpointing, selects affinity modules matching the output being trained, and records gradient coverage. The artifact comparisons provide direct evidence for the expected behavior.

Arm	Artifact evidence	Interpretation
`lora_v2`	294 adapted Linear layers. Nonzero `lora_B`: module 1 stack 184/184 and heads 5/9; module 2 stack 92/92 and heads 5/9. Stack gradient coverage was 368/368 and 184/184 tensors; each head scope was 10/18.	Both affinity stacks and both modules participated.
`ft_v2`	Changed vs. base: module 1 stack 296/296 and heads 10/18; module 2 stack 148/148 and heads 10/18.	Full fine-tuning updated both stacks and both heads.
`lora_legacy`	Nonzero `lora_B`: module 1 heads 5/9; module 1 stack 0/24; module 2 stack and heads 0; confidence stack 0.	The legacy run reproduces the heads-only behavior. Its no-gradient warnings were marked expected for legacy mode.

The v2 LoRA gradient-coverage record reports gradients for every stack tensor and 10 of 18 trainable tensors in each head scope. The artifact check independently confirms nonzero updates in the stack and five of nine adapted head linears per affinity module. The legacy trace emitted PyTorch's "None of the inputs have requires_grad=True" checkpoint warning and explicitly reported zero coverage in its affinity stack.

This is a strong smoke-level confirmation of the documented mitigation. It does not establish that every head tensor must receive gradients; the check here is that the intended stack and affinity-module scopes are active and their artifacts changed.

4. Training behavior

Losses below are the values recorded after each epoch. These short runs should not be compared as a performance ranking: the legacy arm uses different trainer/output semantics, and the arms have very few training examples.

Arm	Epoch 1 train loss	Epoch 1 val loss	Epoch 2 train loss	Epoch 2 val loss
`lora_v2`	1.87793	1.83996	1.71011	1.82152
`lora_legacy`	2.09044	1.81807	2.12949	1.81082
`ft_v2`	1.90277	1.82832	1.58361	1.86374

V2 LoRA training and validation losses both decreased slightly over the two epochs. Full fine-tuning's training loss fell while its validation loss rose from 1.82832 to 1.86374, a warning sign of validation regression at epoch 2 in this tiny run. The legacy arm's training loss rose while its validation loss improved slightly. None of these two-epoch trajectories should be interpreted as a stable generalization estimate.

5. Runtime and memory

Arm	Wall time	Peak GPU memory
`lora_v2`	353 s	35,521 MiB
`lora_legacy`	42 s	5,125 MiB
`ft_v2`	103 s	8,901 MiB

The higher v2-LoRA cost is consistent with the broader active target scope: it adapts 294 linears across both affinity modules, versus 102 legacy adapters whose stack and some other targets received no gradients. These are measurements from a single short smoke job, not throughput estimates for full datasets.

6. Rescoring check

The job selected 20 validation rows. Two structure stems occurred twice, leaving **18 unique input structures**. The resulting vanilla and `lora_v2` CSVs each contain those same 18 IDs, and all 18 rows in each file have `validation_status=SUCCESS`.

Check	Result
Selected validation rows	20
Unique structure IDs / input PDBs	18
Vanilla successful rows	18 / 18
V2-LoRA successful rows	18 / 18
Shared successful rows	18
Mean absolute `affinity_pred` difference, v2 LoRA vs vanilla	0.564
Median / maximum absolute difference	0.599 / 0.750

Three rows in each score CSV carry an `INVALID_ELEMENTS` warning for chlorine (`Cl`) while still having successful validation status. This merits attention in any broader chemistry-focused evaluation, but it did not prevent the smoke test's paired rescoring. The observed prediction differences confirm a changed score path; without labels and an adequately sized held-out evaluation, they do not imply improved accuracy.

7. Caveats and follow-up

1. **Small, clustered subset.** The 64 training rows span only two assay groups and the 32 validation rows span three. Two epochs and this subset validate execution and parameter movement, not generalization.
2. **Censoring was not exercised.** The log says none of the 64 training rows had `is_censored=1`; the selected censored-Huber loss therefore reduced to the uncensored loss. This test does not validate right-censored examples or their handling.
3. **Dirty source tree.** The run records commit `1b7e973...` and `boltz_git_dirty=true`. Preserve the exact patch or rerun from a clean, pinned revision before treating this as a reproducible benchmark.
4. **Duplicate structure names.** The rescore subset has 20 CSV rows but only 18 unique structure stems. Future checks should report the unique input count explicitly (or deduplicate rows) so the requested and actual pose counts are unambiguous.
5. **Compatibility warning.** The log notes that the checkpoint was produced with PyTorch Lightning `2.5.0.post0`, newer than the environment's `2.5.0`. Loading and scoring completed, but matching the checkpoint's dependency version is preferable for controlled runs.

Recommended next step: run a larger v2 validation/training job from a clean pinned revision, with enough independent assay groups and censored examples if censor handling is in scope. Use a new `_v2` arm tag, retain the generated `meta.json` provenance, and compare held-out metrics against vanilla rather than relying on training loss or score shifts alone.

8. Sources

- Smoke-test log: `trainer_v2_smoke_27998524.log`
- Migration and mitigation notes: `MIGRATION_TRAINER_V2.md`
- Smoke-test job script: `trainer_v2_smoke.slurm`
- Saved run artifacts: `/scratch/maom_root/maom/limcaoco/trainer_v2_smoke_27998524`